

LUODIAN, China

WMO: 579160

Lat: 25.4286N

Lon: 106.7638E

Elev: 1447

StdP: 13.94

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	37.9	40.3	27.0	22.1	46.0	30.2	25.6	47.1	8.2	58.7	7.2	56.4	1.6	140	0.278

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.4	95.2	77.6	93.5	77.5	91.7	77.0	80.4	90.8	79.6	89.5	78.8	88.2	3.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.9	153.1	85.0	77.1	149.1	84.2	76.4	145.5	83.3	45.6	91.2	44.6	89.8	43.8	88.4	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.7	6.4	5.3	DB	34.1	98.9	2.3	1.6	32.4	100.0	31.1	100.9	29.8	101.8	28.1	103.0	
			WB	32.7	82.2	2.4	1.0	30.9	82.9	29.5	83.5	28.1	84.1	26.3	84.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	69.3	51.7	56.3	63.3	71.9	76.6	79.7	82.0	81.8	77.9	71.1	63.4	54.6
	DBStd	12.05	7.00	8.64	7.95	7.23	5.23	3.79	2.87	3.16	4.59	5.36	5.93	5.35
	HDD50	108	62	27	2	0	0	0	0	0	0	0	1	16
	HDD65	1273	416	268	132	20	0	0	0	0	0	11	100	326
	CDD50	7139	114	205	415	658	825	891	993	986	838	655	402	157
	CDD65	2831	3	26	80	228	361	441	528	521	388	201	51	3
	CDH74	24475	23	228	577	1836	3039	3927	5288	5008	3162	1122	243	22
	CDH80	9587	2	71	197	735	1178	1460	2176	2158	1246	323	38	3
Wind	WSAvg	1.8	1.7	2.1	2.2	2.2	1.9	1.7	1.6	1.6	1.6	1.4	1.4	1.6
Precipitation	PrecAvg	45.3	0.6	0.7	1.2	3.7	8.2	8.3	7.4	6.3	3.0	3.4	1.8	0.7
	PrecMax	64.0	1.7	2.0	5.0	9.9	22.2	16.5	18.1	11.5	7.4	10.1	6.6	2.6
	PrecMin	30.5	0.0	0.1	0.0	0.7	2.4	2.4	0.9	2.0	1.0	0.6	0.0	0.0
	PrecStd	9.0	0.4	0.4	1.1	2.2	4.2	3.5	4.3	2.6	1.7	2.6	1.5	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.9	87.9	91.3	96.8	96.3	95.3	96.6	96.7	95.8	89.8	84.5	77.3
		MCWB	63.3	65.9	69.2	71.7	76.6	79.5	79.4	78.3	76.9	72.5	69.8	64.4
	2%	DB	70.9	81.0	85.5	92.1	92.6	92.6	94.1	94.6	92.5	86.5	80.0	70.1
		MCWB	60.0	65.0	68.9	72.9	76.1	78.6	79.2	78.3	76.2	72.3	67.6	59.6
	5%	DB	65.8	75.0	80.6	88.1	89.7	90.2	92.1	92.6	90.1	83.7	76.6	66.3
		MCWB	56.7	63.0	67.1	72.4	74.9	77.8	78.7	78.1	75.3	71.1	65.8	57.0
	10%	DB	61.4	69.4	75.6	84.0	86.7	87.8	90.0	90.4	87.3	80.5	73.2	62.9
		MCWB	54.7	60.3	64.5	70.8	73.8	77.0	78.2	77.6	74.2	69.9	64.2	55.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.6	67.8	71.7	76.3	79.8	81.5	81.7	81.0	79.5	76.2	71.6	64.8
		MCDB	75.5	82.4	85.3	89.5	91.6	92.2	92.1	91.7	90.6	85.6	79.9	74.7
	2%	WB	60.8	65.2	69.5	74.6	78.0	80.1	80.5	79.9	78.2	74.6	69.7	61.5
		MCDB	69.8	78.1	82.2	87.1	89.0	89.8	90.7	90.4	88.3	83.0	76.8	68.1
	5%	WB	58.3	63.1	67.7	73.2	76.7	79.1	79.6	79.1	77.1	73.1	67.8	59.1
		MCDB	64.1	74.1	78.8	84.5	86.6	87.6	89.0	89.4	86.4	80.6	74.9	64.4
	10%	WB	55.6	60.9	65.8	71.8	75.4	78.1	78.8	78.2	76.1	71.5	65.7	56.5
		MCDB	60.3	69.0	74.6	81.9	83.7	85.4	87.5	87.6	84.0	77.6	71.2	61.5
Mean Daily Temperature Range	5% DB	MDBR	11.5	13.8	14.8	16.1	15.4	13.5	14.4	15.8	16.0	13.7	14.2	12.9
		MCDBR	21.7	25.8	23.4	22.7	21.0	17.9	18.6	19.9	20.6	19.5	20.3	21.6
		MCWBR	10.6	10.3	8.5	7.2	7.0	5.8	5.7	5.5	5.9	6.5	8.7	11.0
	5% WB	MCDBR	19.4	23.7	21.9	20.1	17.0	15.1	16.1	17.6	17.0	15.9	17.1	18.2
MCWBR		9.9	9.7	8.7	7.5	6.6	5.5	5.6	5.6	5.3	6.0	7.9	10.0	
Clear-Sky Solar Irradiance	taub	0.619	0.621	0.746	0.708	0.584	0.566	0.525	0.579	0.596	0.680	0.622	0.648	
	taud	1.612	1.644	1.453	1.570	1.893	1.962	2.082	1.912	1.837	1.600	1.656	1.572	
	Ebn at Noon	197	212	194	208	236	238	248	235	226	195	195	182	
	Edh at Noon	74	76	97	88	64	59	52	62	65	79	70	74	
All-Sky Solar Radiation	RadAvg	569	822	1004	1279	1333	1217	1481	1441	1240	895	831	622	
	RadStd	144	170	134	162	153	160	106	104	123	126	124	103	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	+0.77	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	N/A	+0.74	N/A	N/A	N/A	N/A	N/A	+88

Nomenclature: See separate page